



→ Inside Readers

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Agenda

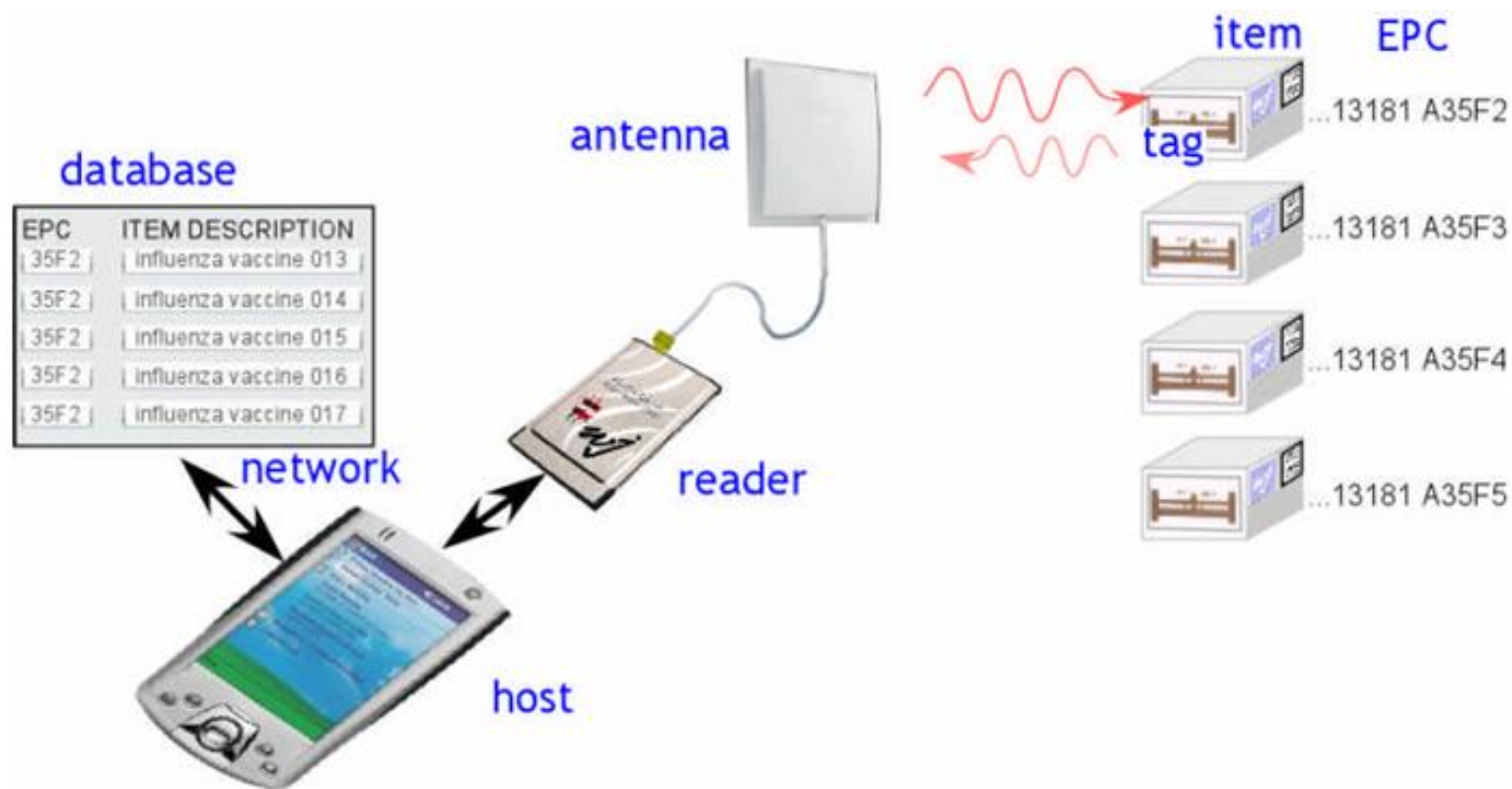


- **Reader introduction**
- **Reader performance**
- **Reader architecture**
 - Antenna
 - Transmitter
 - Receiver
- **Gen 2 dense reader mode technology**
- **Q&A**





RFID System





Reader Layers



Content of Reader/Host Exchanges (abstract syntax)

Reader Layer

Message formatting

- Message framing
- Message syntax
- Error information

Messaging Layer

O/S-provided network facility

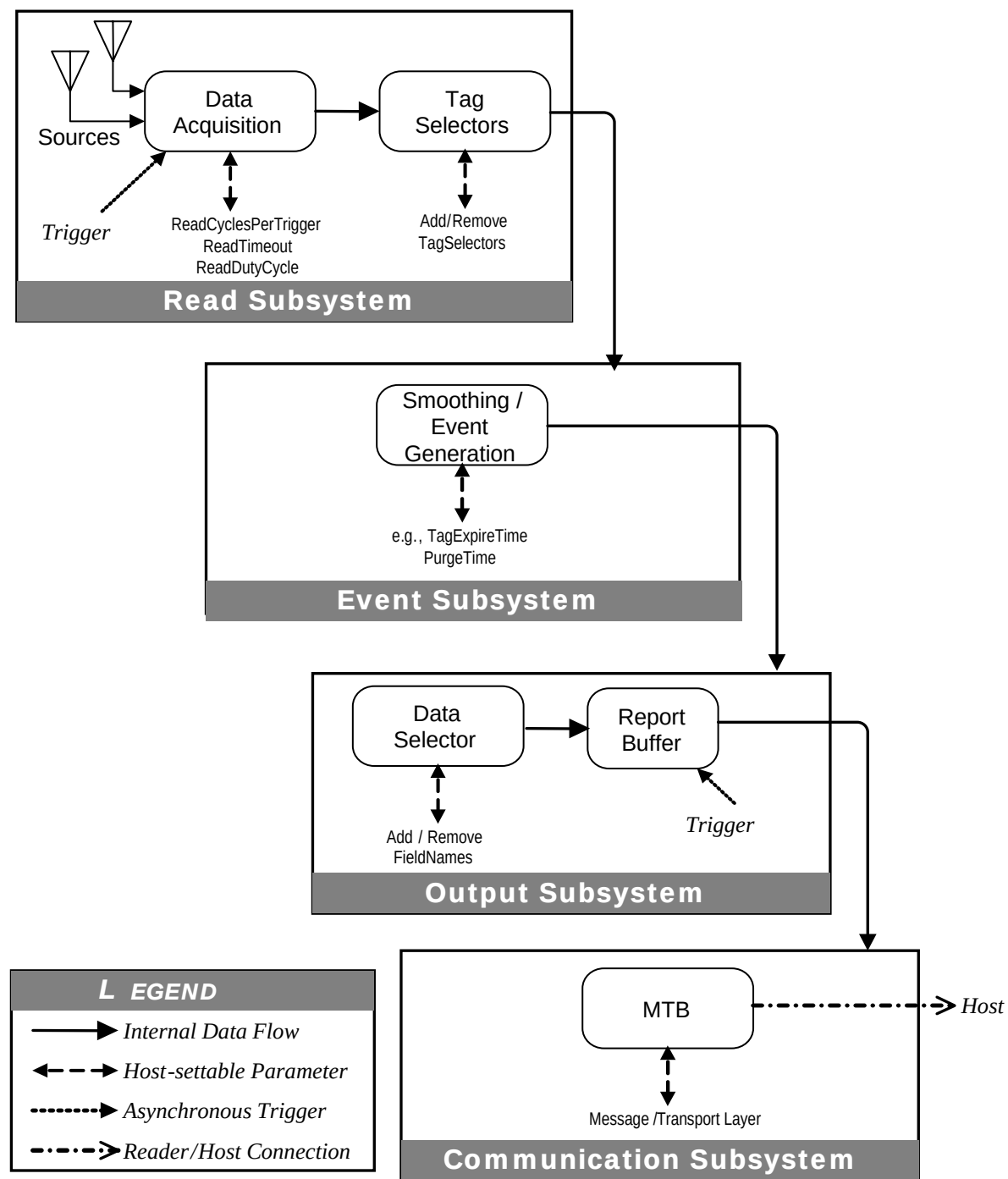
Transport Layer

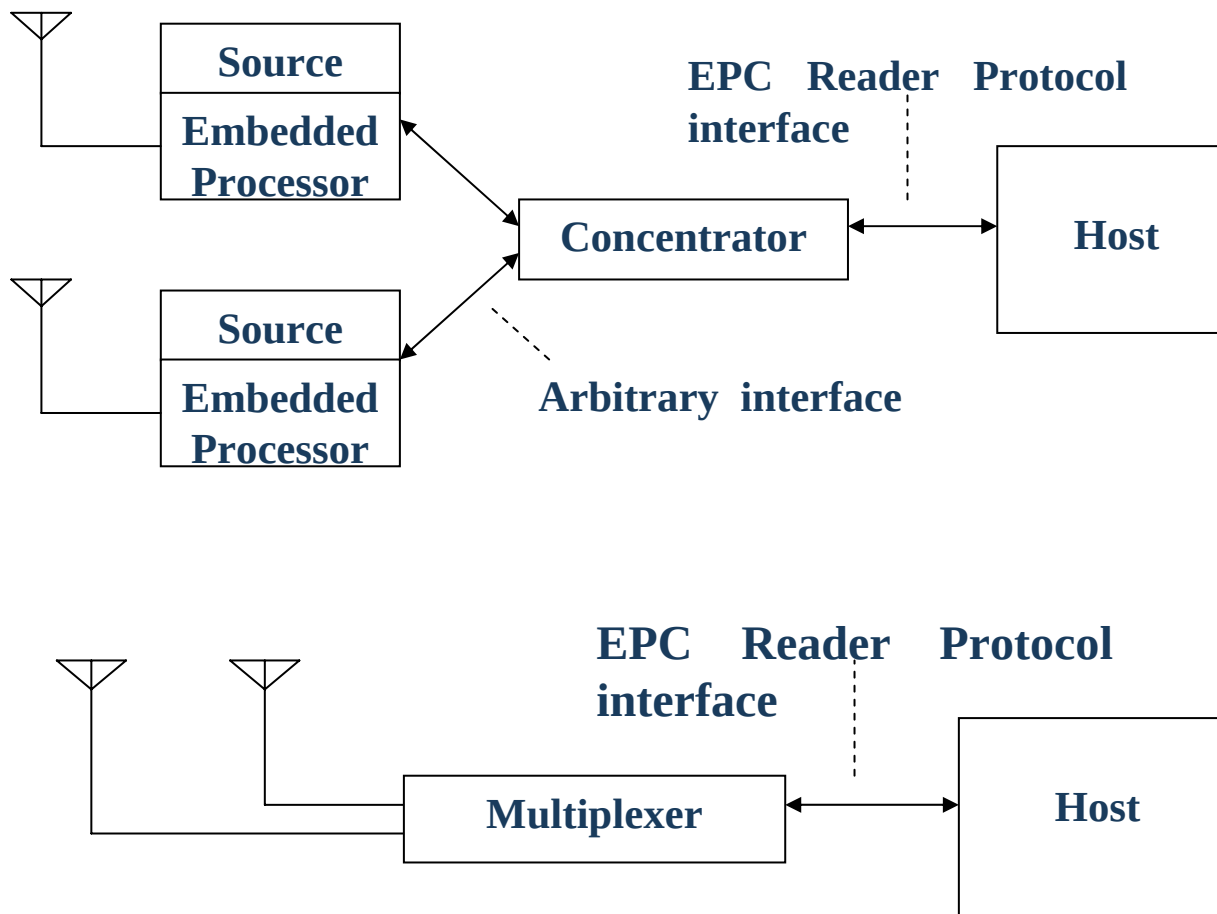
Each pair is called a Messaging /Transport Binding (MTB)



Reader Pipeline

- Read subsystem
- Event subsystem
- Output subsystem
- Communication subsystem







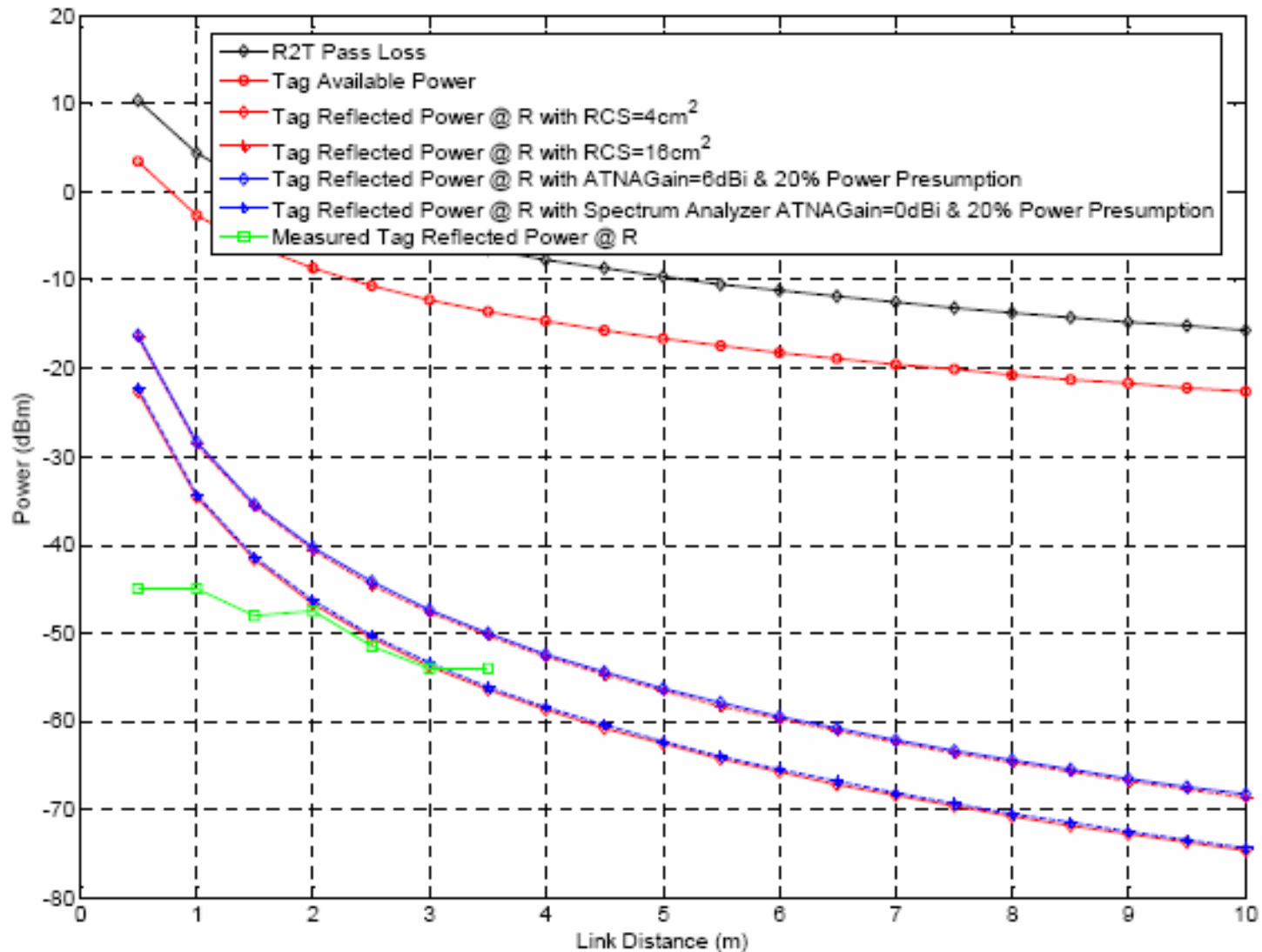
- **Interface to tags**
 - Application point of view
 - Protocols supported
 - Frequency
 - Read range
 - Singulate rate
 - RF characteristics
 - RF output power
 - Out of band emission
 - Noise mask
 - Envelope
 - Sensitivity

- **LF: 125K, 134K**
 - ISO 18000-2
 - ISO 14423
- **HF: 13.56M**
 - ISO 18000-3
 - ISO 15693
 - ISO 14443 A/B
- **UHF 860-960 MHz**
 - EPC Class 0, EPC Class 1, EPC Gen2
 - ISO 18000-6 A/B/C
 - Complaint to regional radio regulation (FCC, ETSI, etc.)
- **MW 2.45GHz**
 - ISO 18000-4

- **Tag**
 - Enough power to operate
 - Correctly decode R->T signaling
 - Backscatter enough power
- **Reader**
 - Transmit enough power: RF output
 - Get enough power from tag backscattering: Sensitivity
 - Correctly decode T->R signaling: Signal noise ratio



Read Range



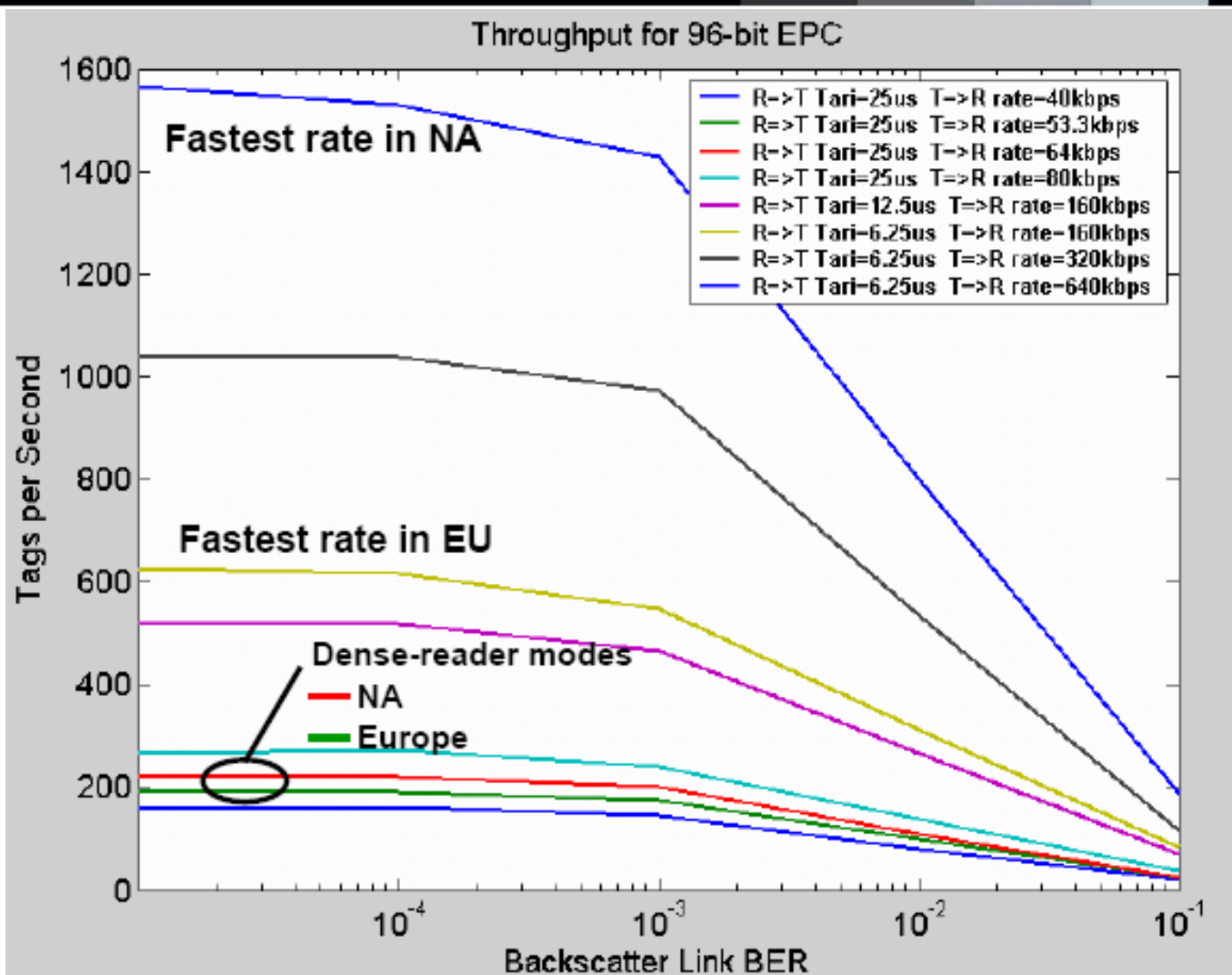


Singulate Rate



- **Number of tags being singulated in one second**
- **Determined by**
 - Down link speed
 - Up link speed
 - Channel signal noise ratio
 - Anti-collision algorithm
 - Total number of tags being powered







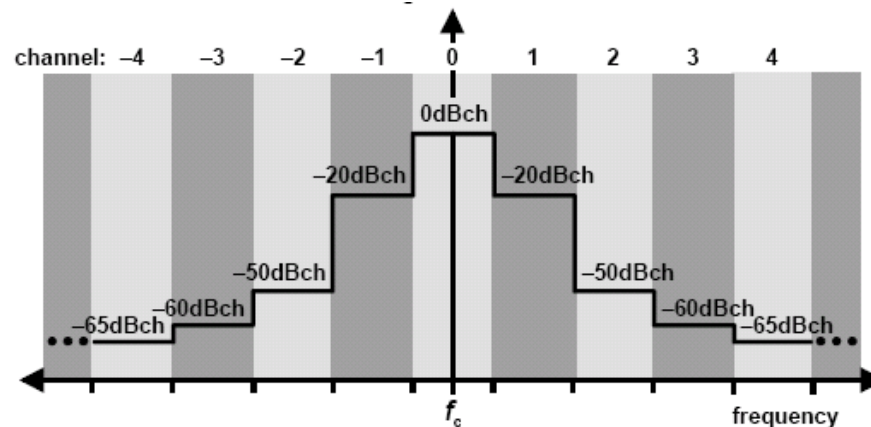
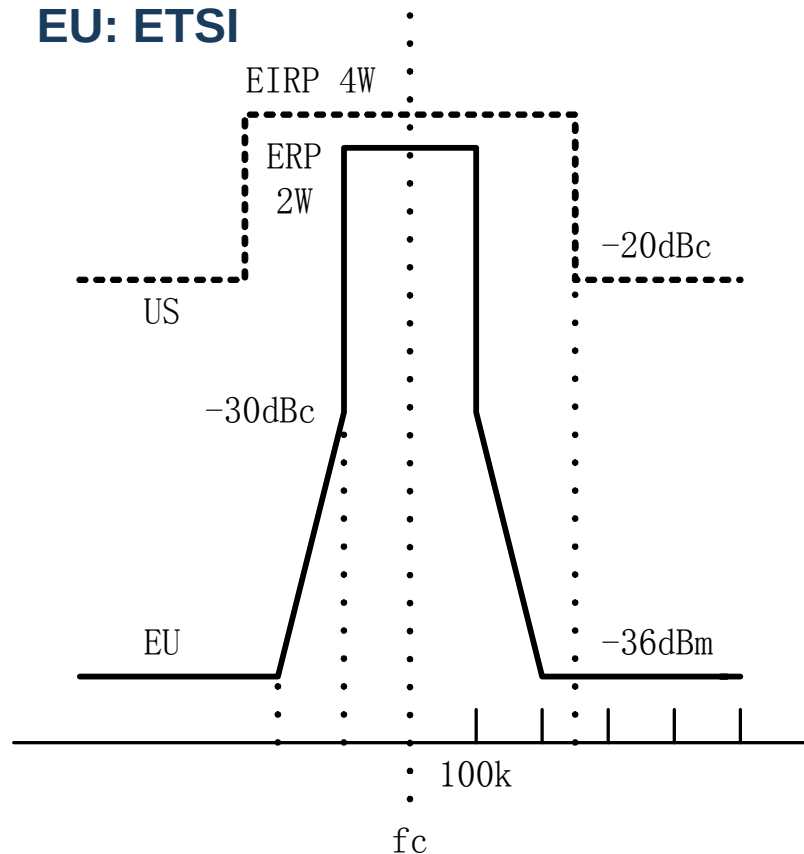
RF Characteristics



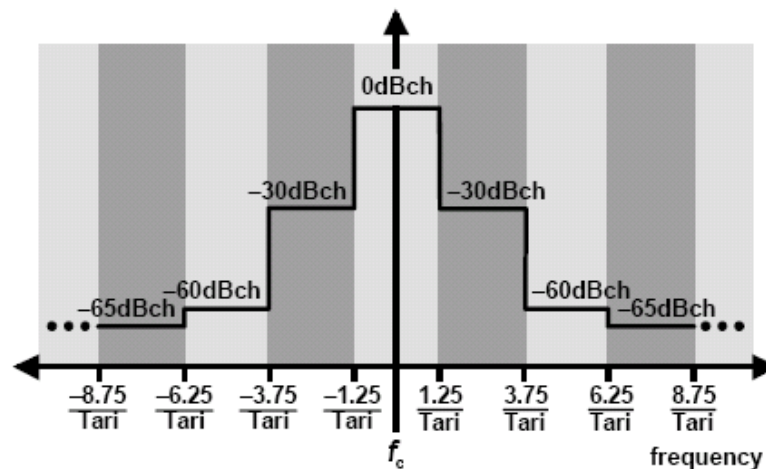
Output power

US: FCC

EU: ETSI



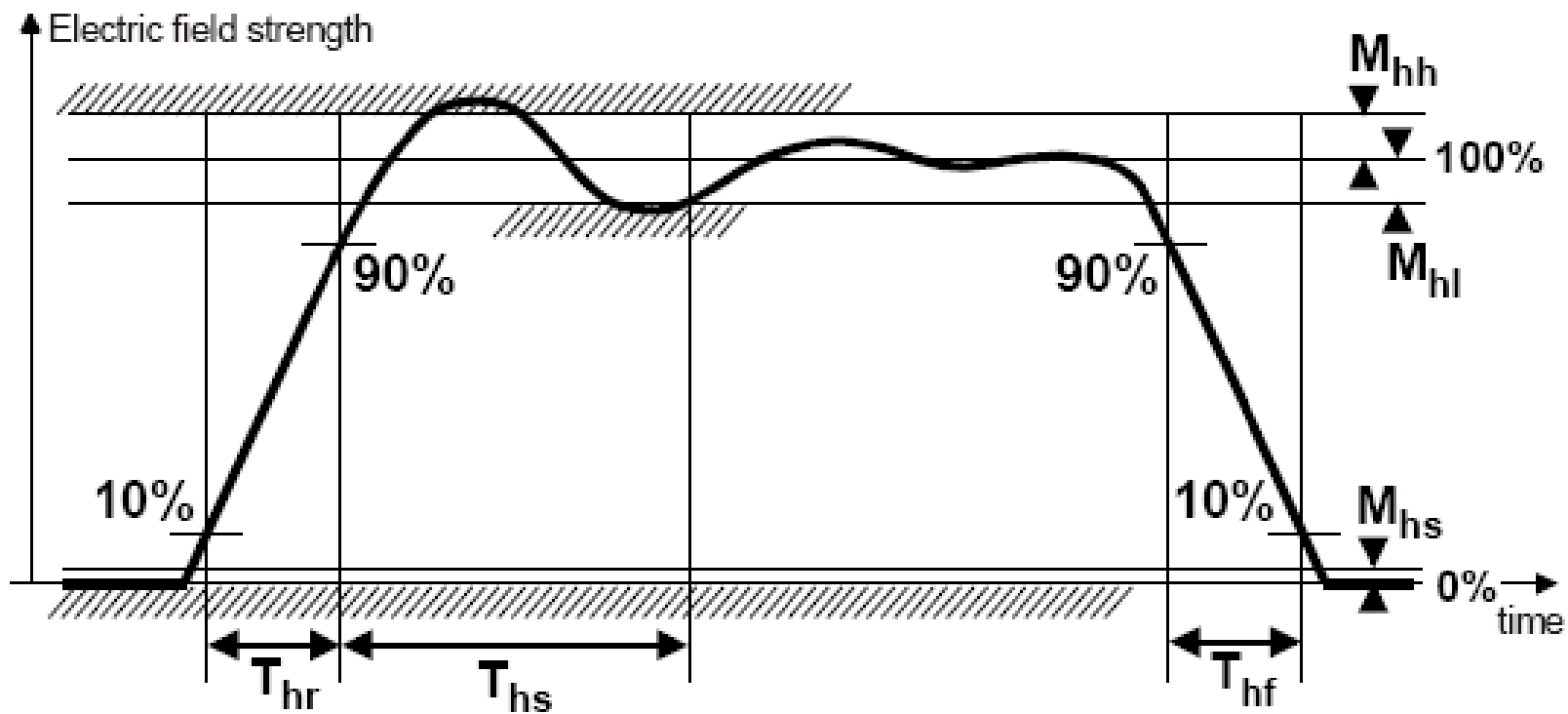
Multiple Readers

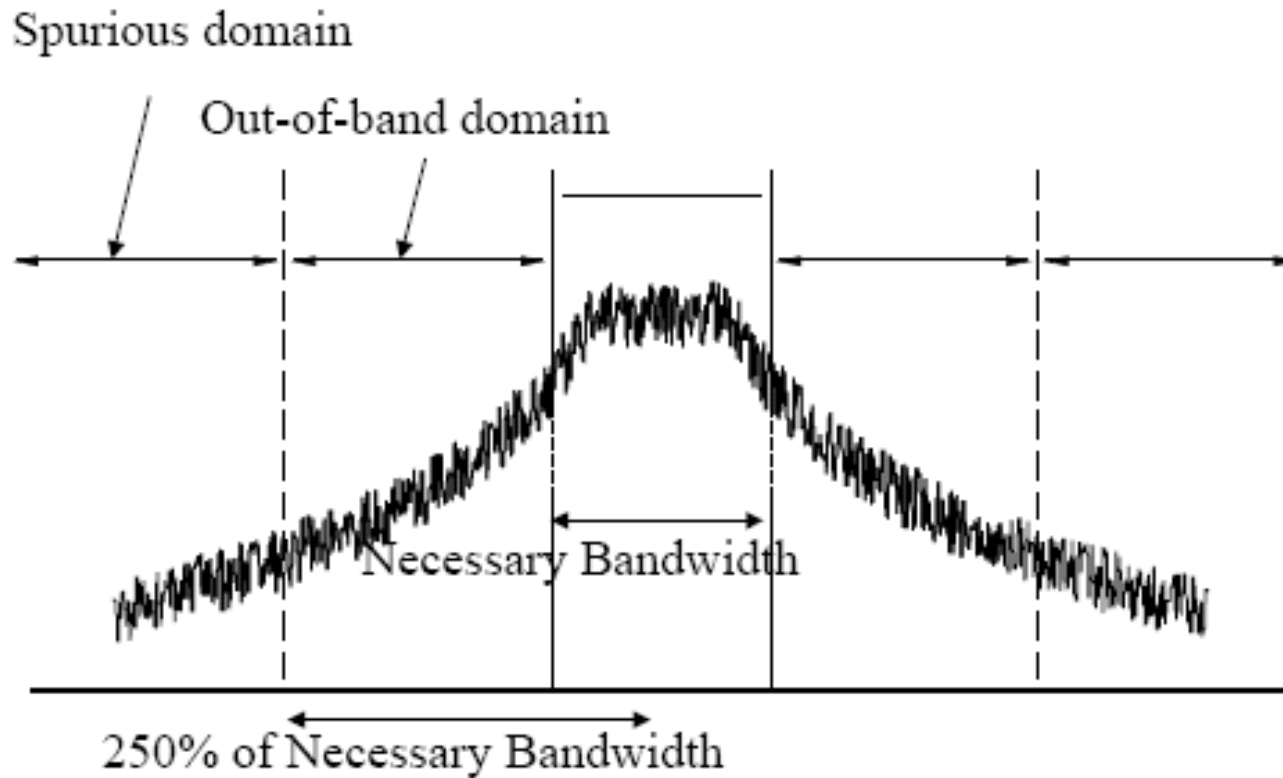


Dense Readers



Envelope





- ETSI: 2nW below 1GHz, 20nw above 1GHz

- **Sensitivity**
 - Minimum backscattered power to be successfully recognized under a defined bit error rate (BER)
 - -70 ~ -85 dbm



Host Interface



- **Transport layer**
 - TCP
 - XML
 - Serial
- **Physical layer**
 - Network
 - LAN
 - WLAN
 - Serial
 - RS232, RS485
 - USB
 - Bluetooth





Reader Architecture

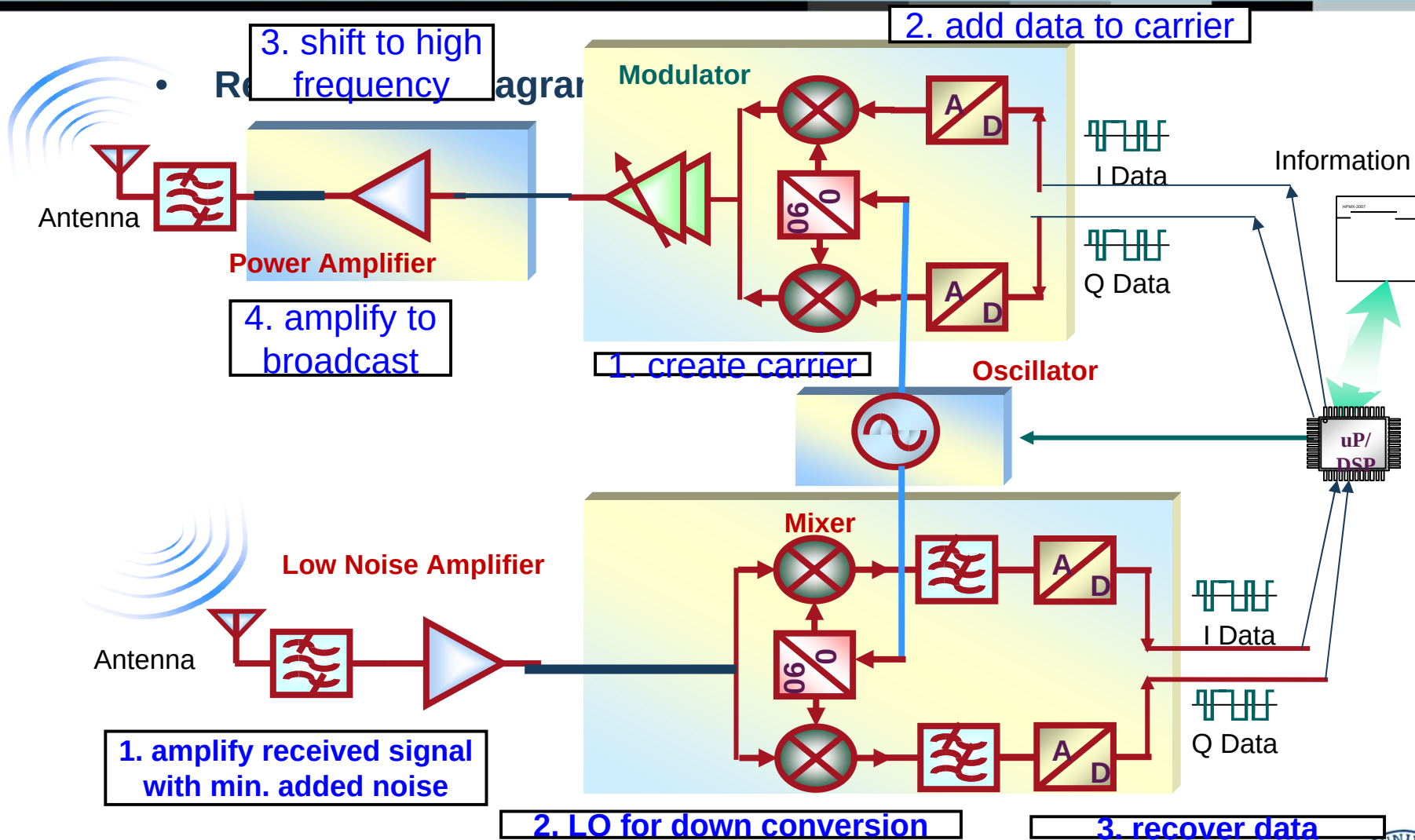


- Reader block diagram
- System behavior model
- Transmitter
- Receiver
- Tx/Rx isolator
- Antenna

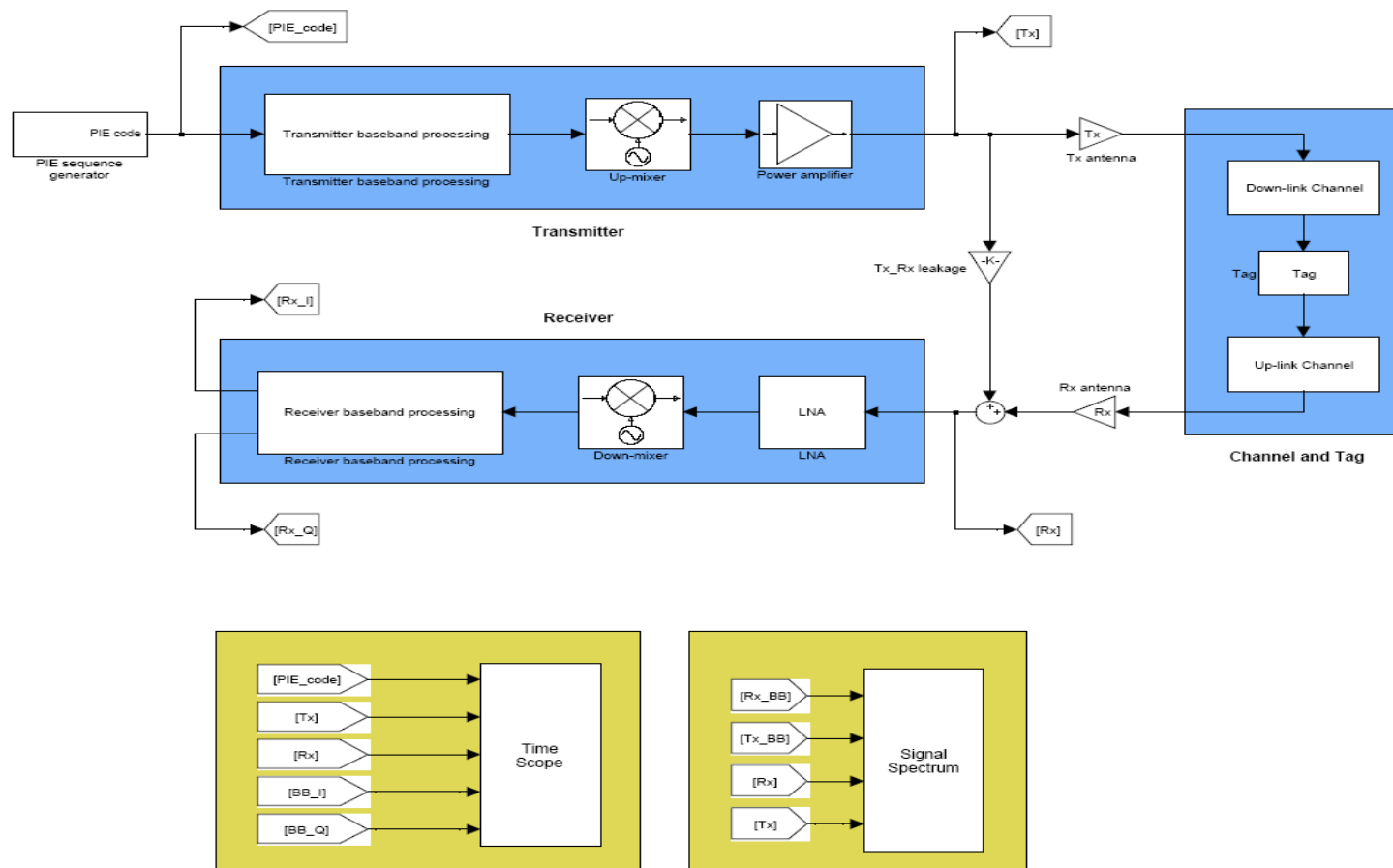




Reader Architecture

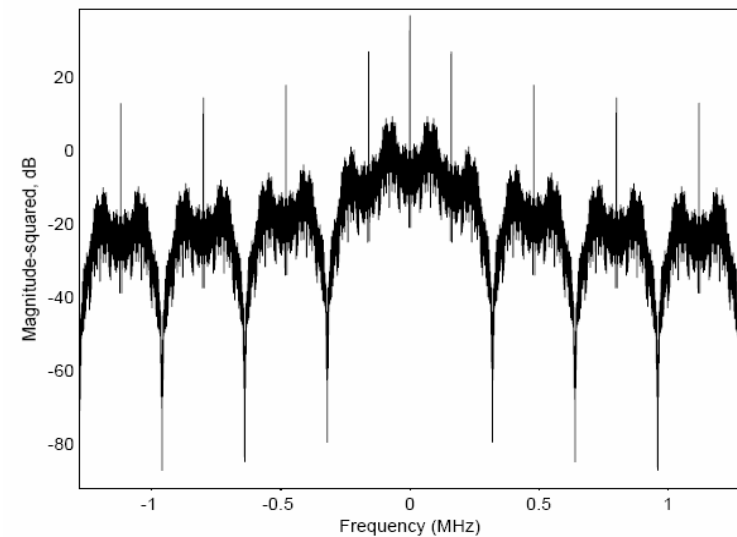
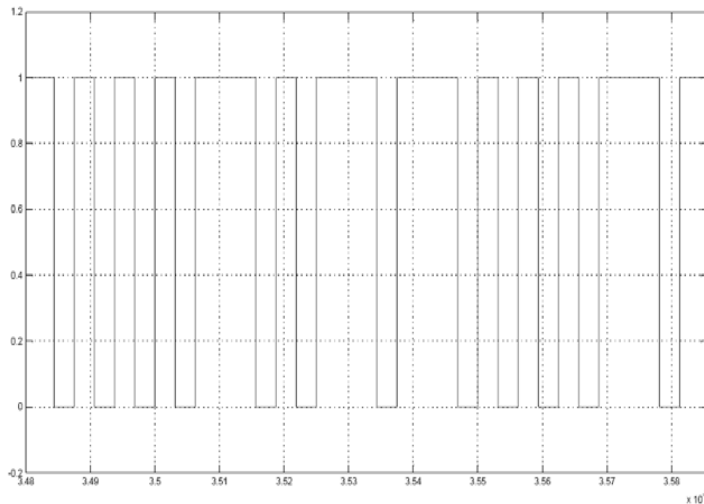


- System behavior model



- **Baseband processing**
 - Protocol processor
 - CRC generator
 - Symbol generator (PIE encoder)

160k PIE encoded signal:



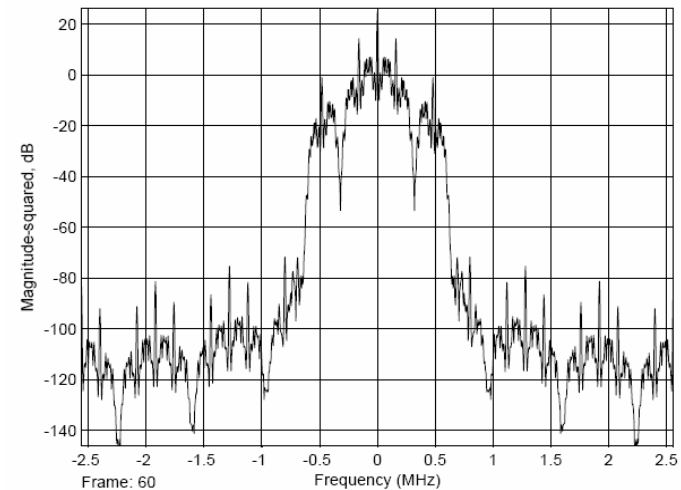
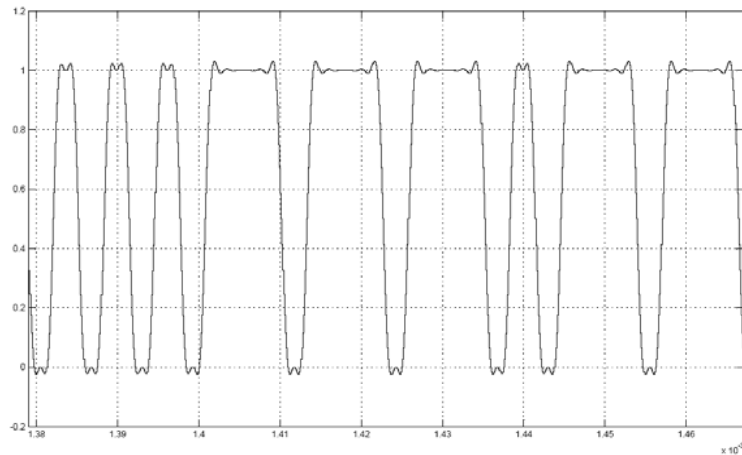


Reader Transmitter



- **Baseband processing**
 - Raised cosine filtering

Raised cosine filtered PIE data (roll off factor = 1, up-sampling rate = 16):



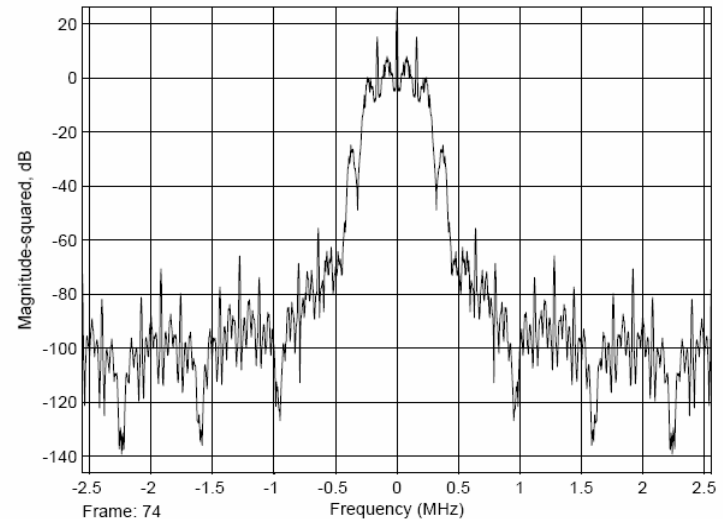
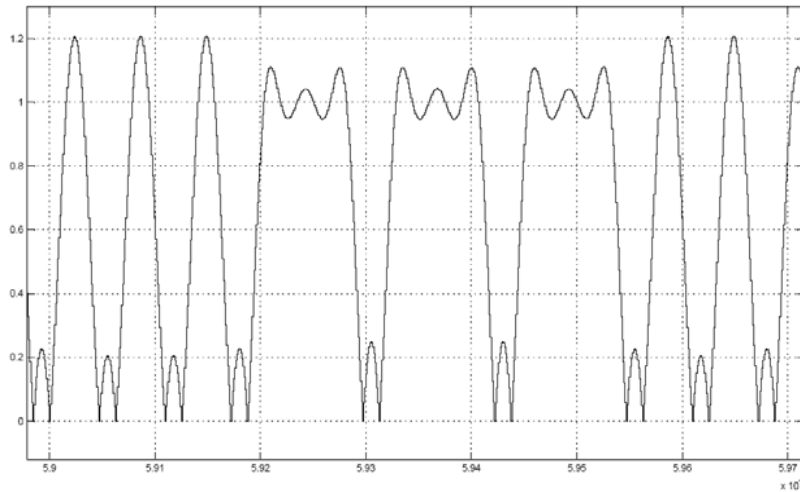


Reader Transmitter



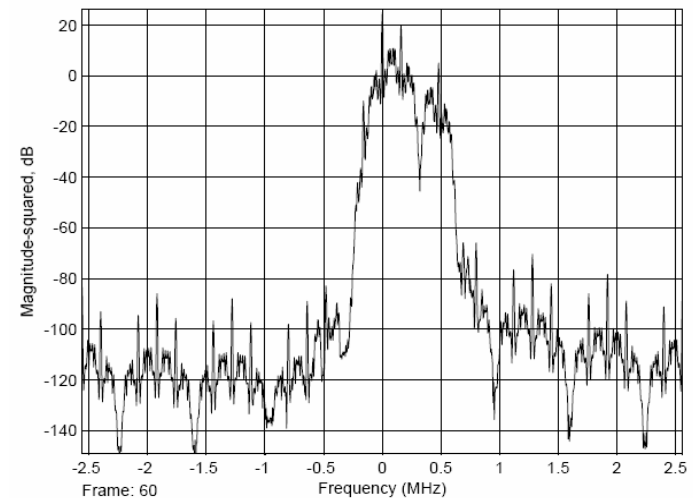
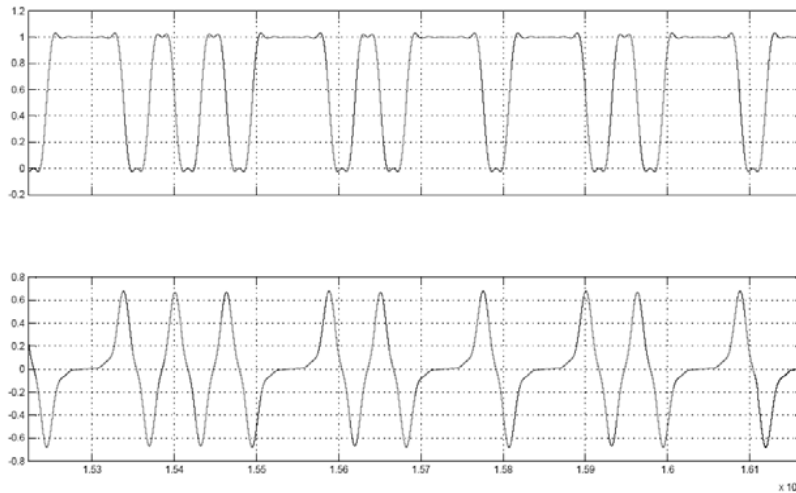
- **Baseband processing**
 - Raised cosine filtering

Raised cosine filtered PIE data (roll off factor = 0.3, up-sampling rate = 16):



- **Baseband processing**
 - Hilbert transform

SSB baseband signal:



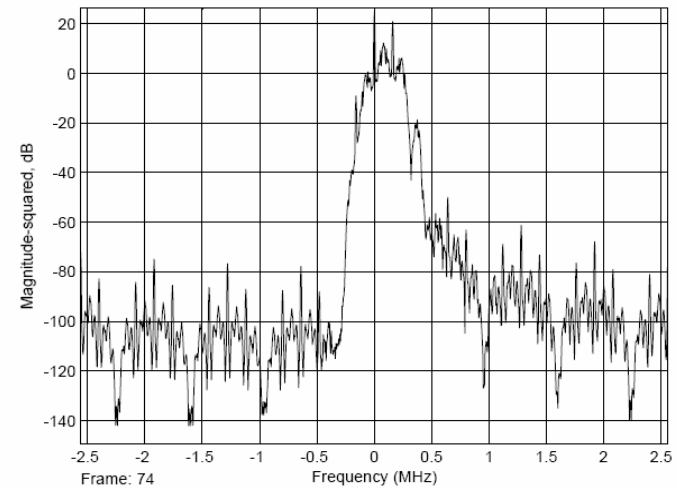
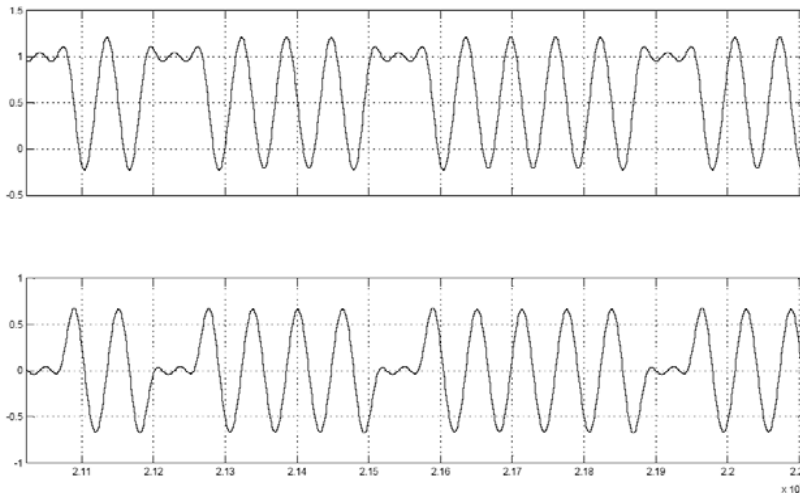


Reader Transmitter



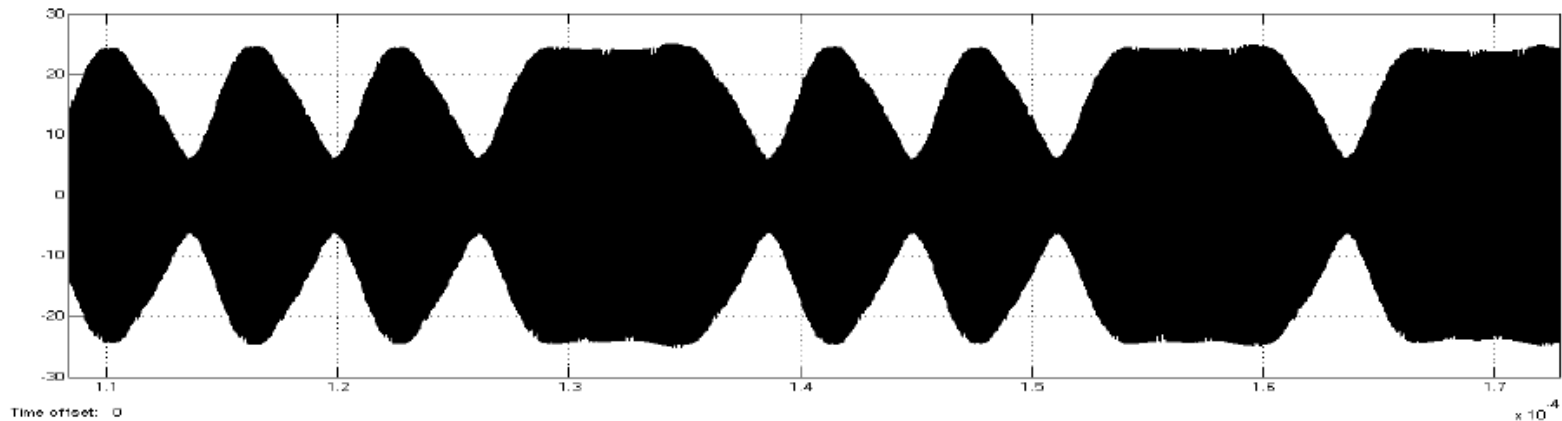
- **Baseband processing**
 - Hilbert transform

SSB baseband signal (roll factor =0.3):



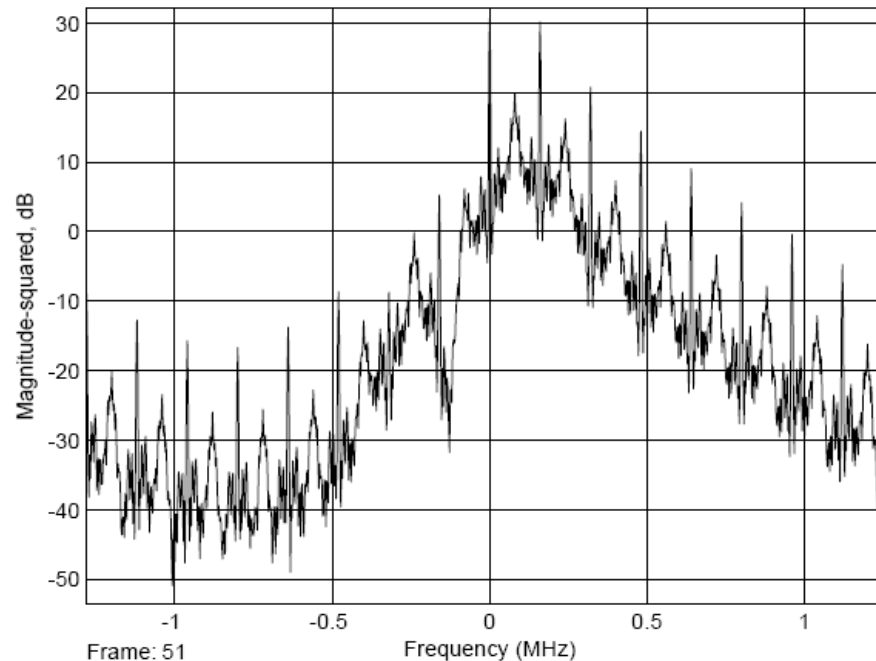
- **Transmitter front-end**
 - Modulation and amplification

ASK modulated SSB signal with noise and distortion (time domain):

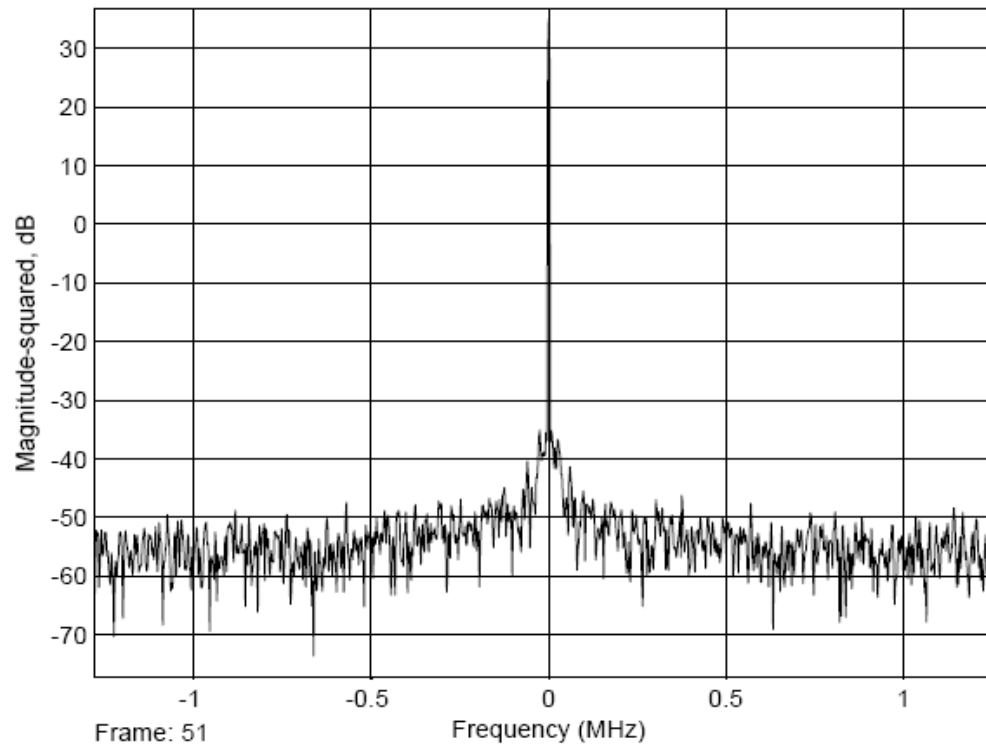


- **Transmitter front-end**
 - Modulation and amplification

ASK modulated SSB signal with noise and distortion (time domain):



- Transmitter front-end
 - Pure carrier with noise



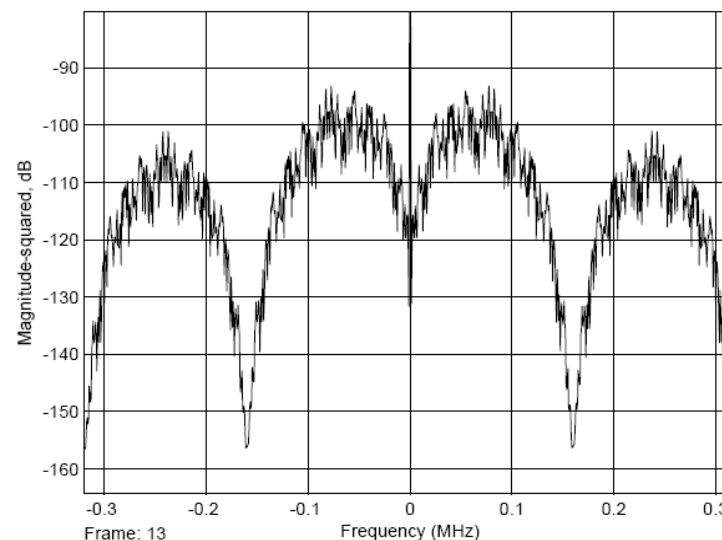
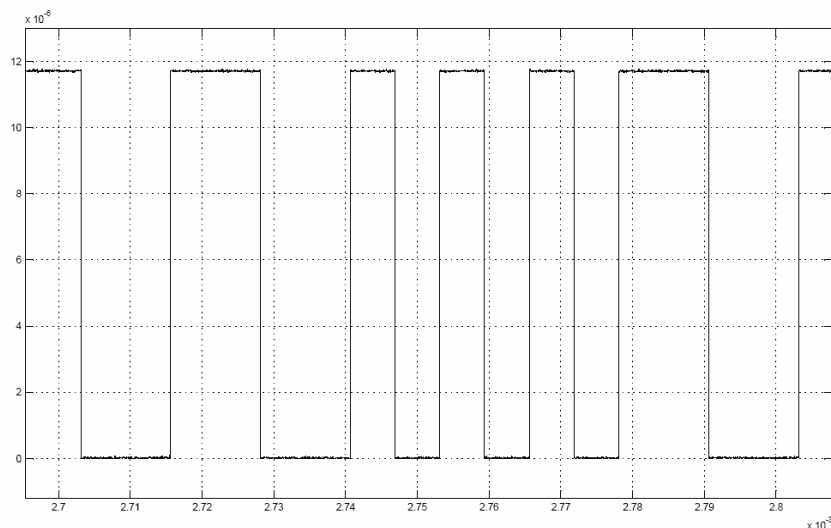


Reader Receiver



- **Backscattered signal**

80KHz FMO coded backscattered baseband equivalent signal
(with path loss and thermal noise):



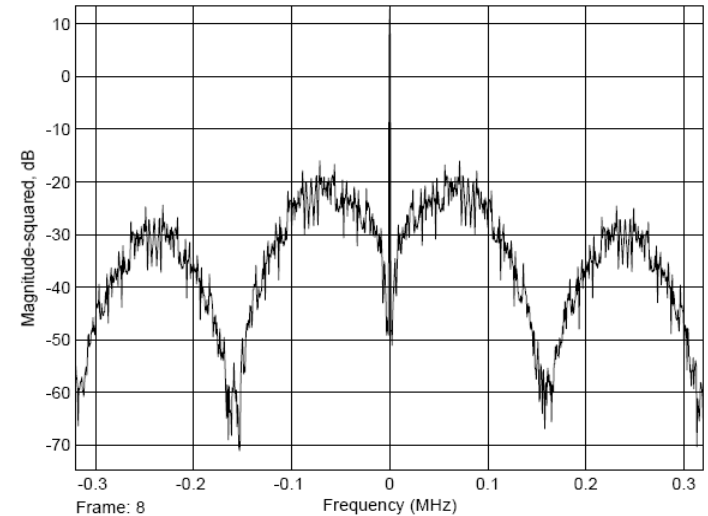
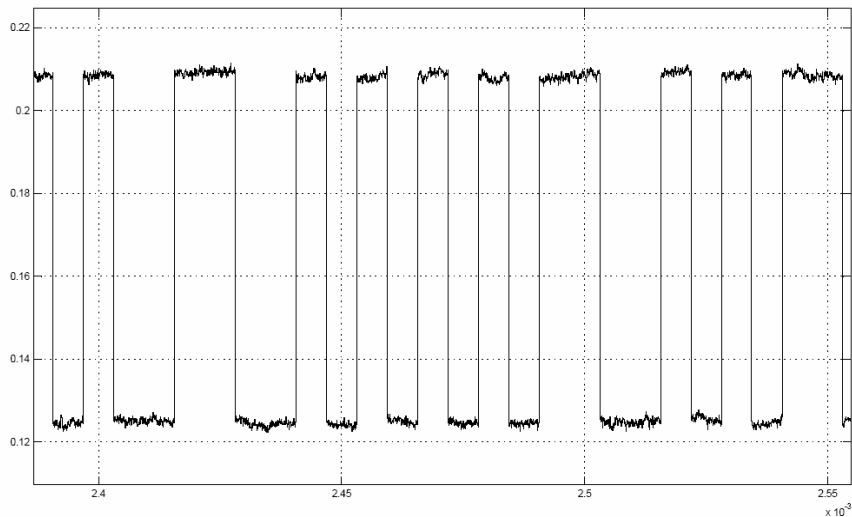


Reader Receiver



- Down-converted backscattering signal

Down-converted signal (Tx-Rx isolation = 50dB):





Reader Receiver

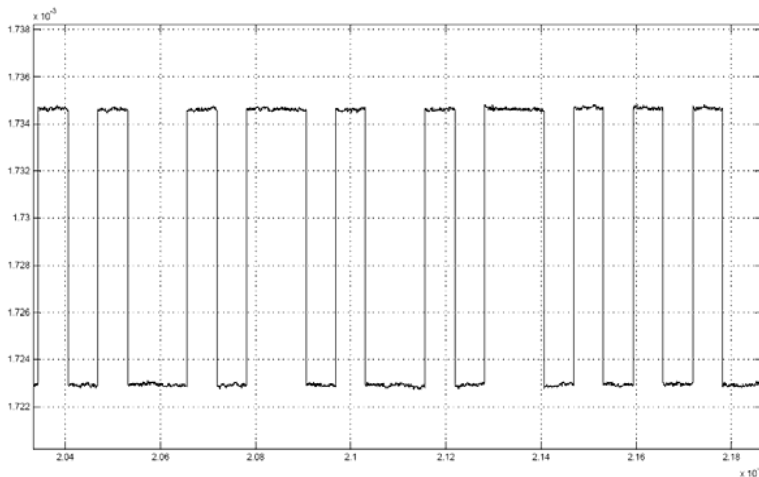


- **Down-converted backscattering signal**

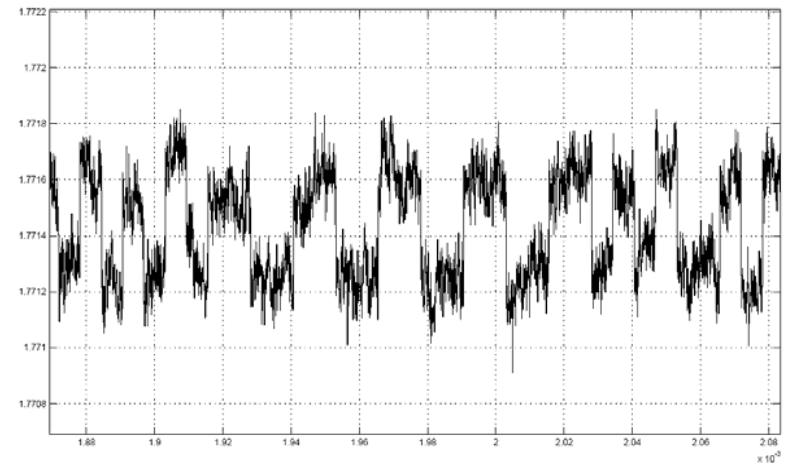
Down-converted signal (Tx-Rx isolation = 30dB)

The signal is desensitized greatly and is easily corrupted by the noise in frontend circuit

Signal after isolator:



Signal after LNA-Mixer:



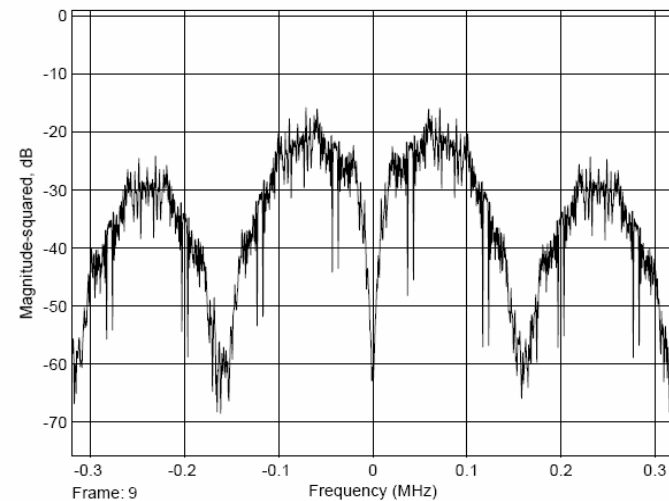
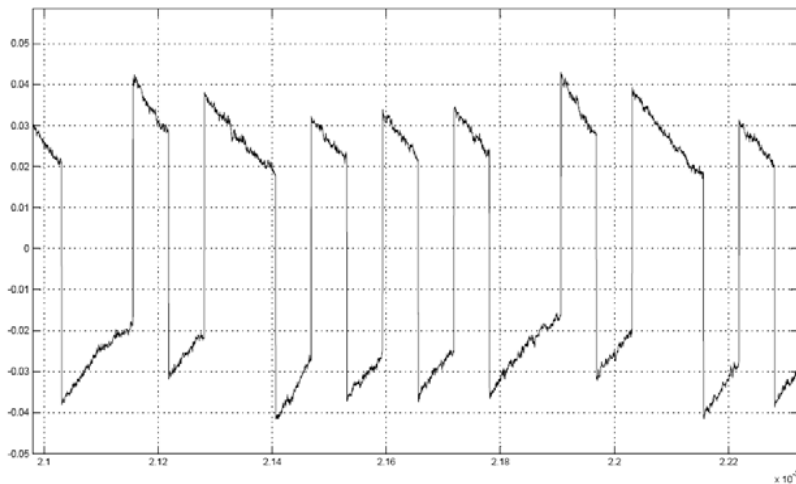


Reader Receiver



- Band pass filtering

AC coupling (10kHz high pass pole):



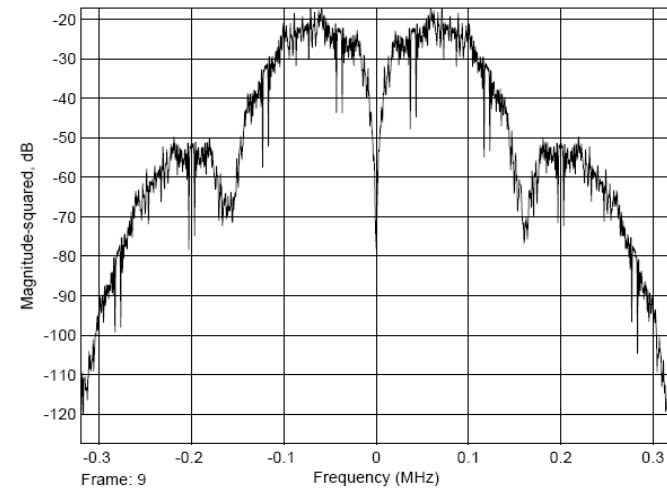
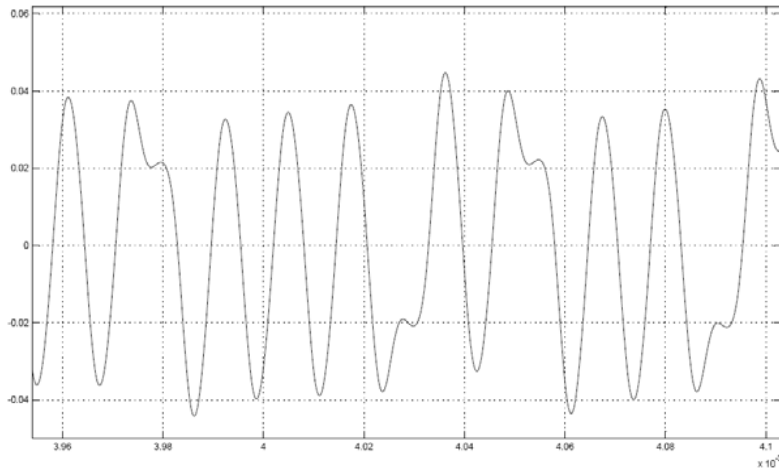


Reader Receiver



- Bandpass filtering

LPF (160kHz -3dB bandwidth):



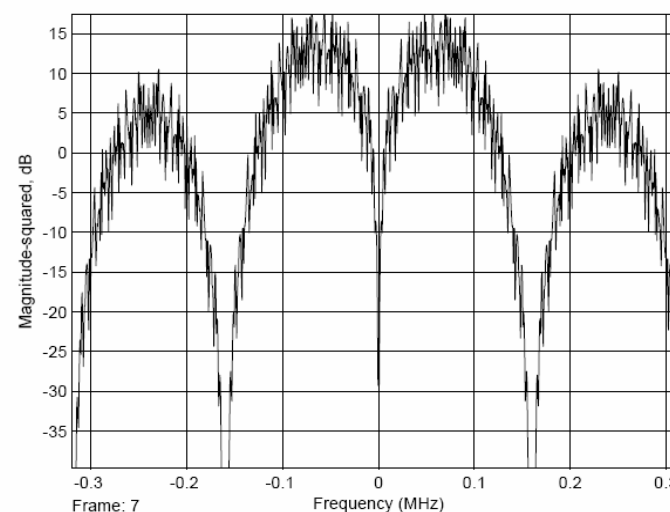
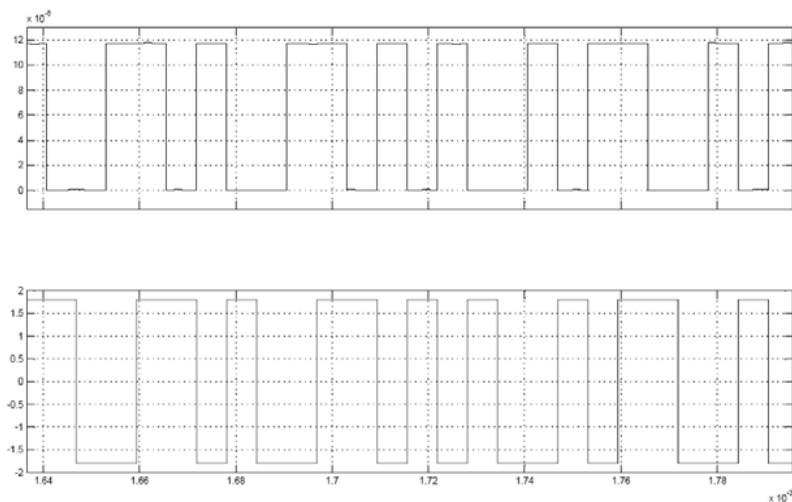


Reader Receiver



- Signal scaling before ADC

VGA scaled signal (compared with the received signal at antenna):



- **In the coherent demodulator, 20-30dB isolation is not enough for 1W output power level**
 - Multiple circulator (high cost and large volume);
 - Dual antenna (complex structure, high cost and large volume);
 - Active antenna (power hungry);
 - Coupler (undesired coupling loss and moderate isolation);
 - Seeking for better solutions...



Gen 2 Dense Reader Mode Technology



- Overview
- Reader to reader interference
- Reader to tag interference
- Dense reader mode technology
- Section A/B





Why Gen2?

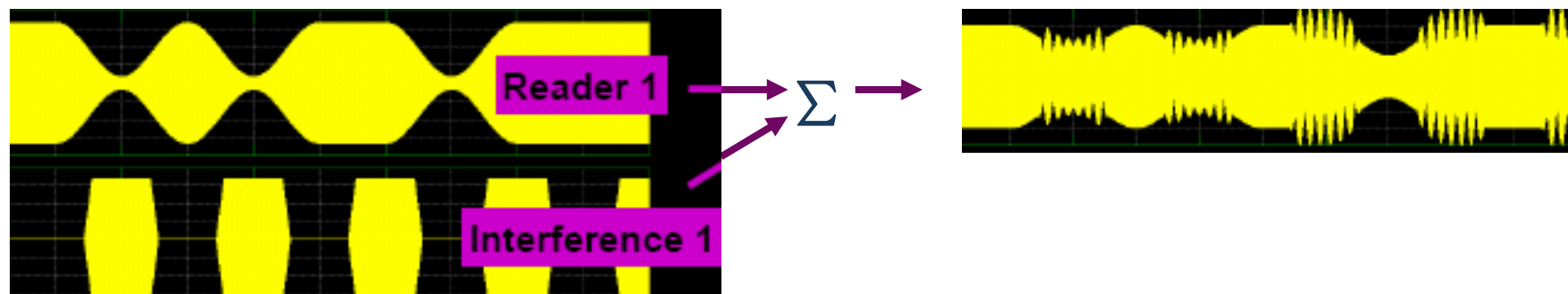
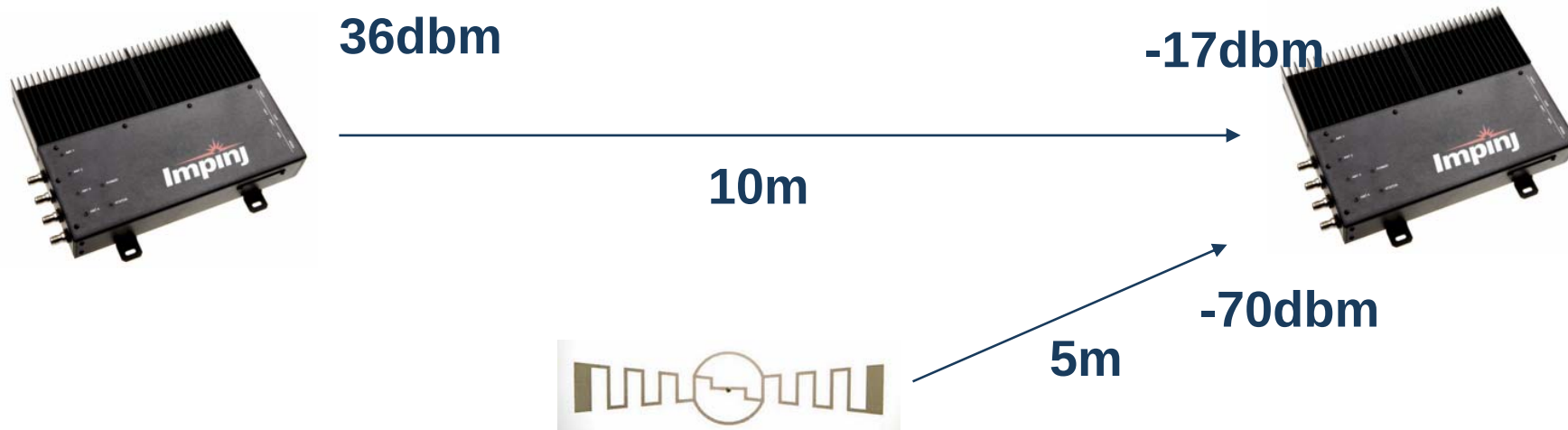


Features	Class 0	Class 1	Gen2
WW regulatory compliance		✓	✓
Memory access control & locking	✓		✓
Optional user memory			✓
Multi-source availability	✓		✓
Reads > 500 tags / second	✓		✓
Kill security			✓
Low cost	✓		✓
Dense-reader operation		✓	✓
Field-rewriteable memory			✓
Industry certification plan			✓

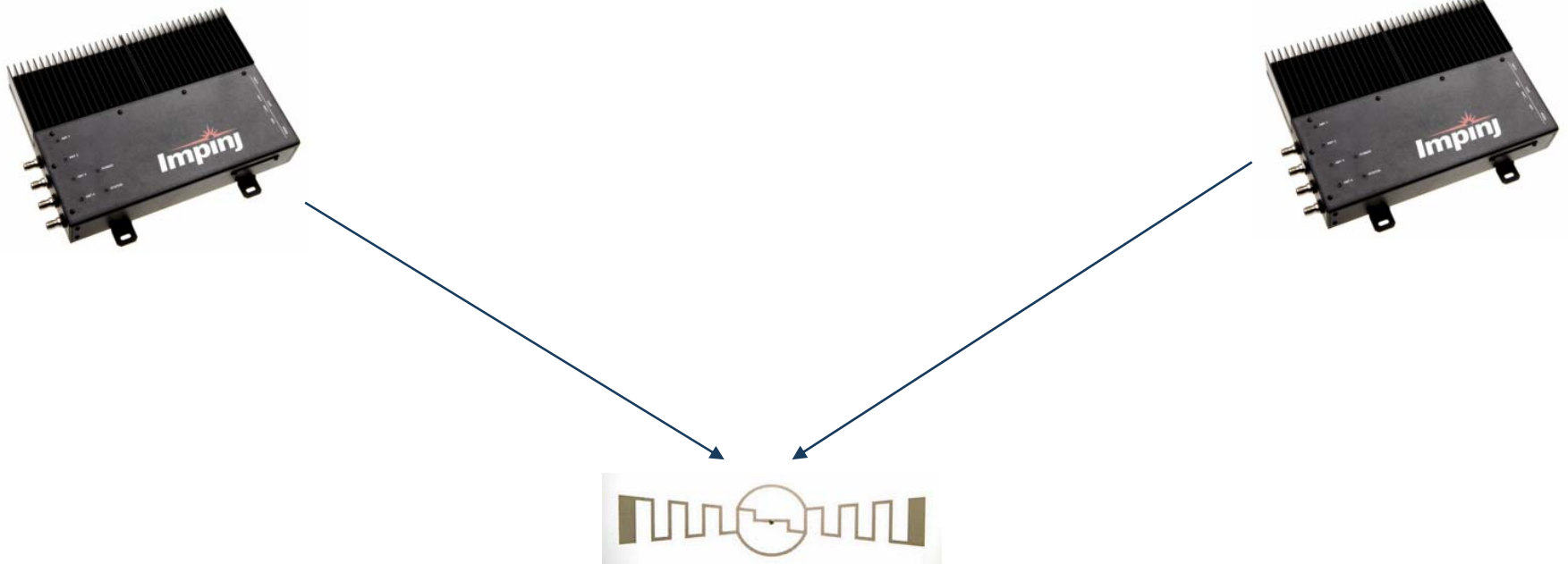


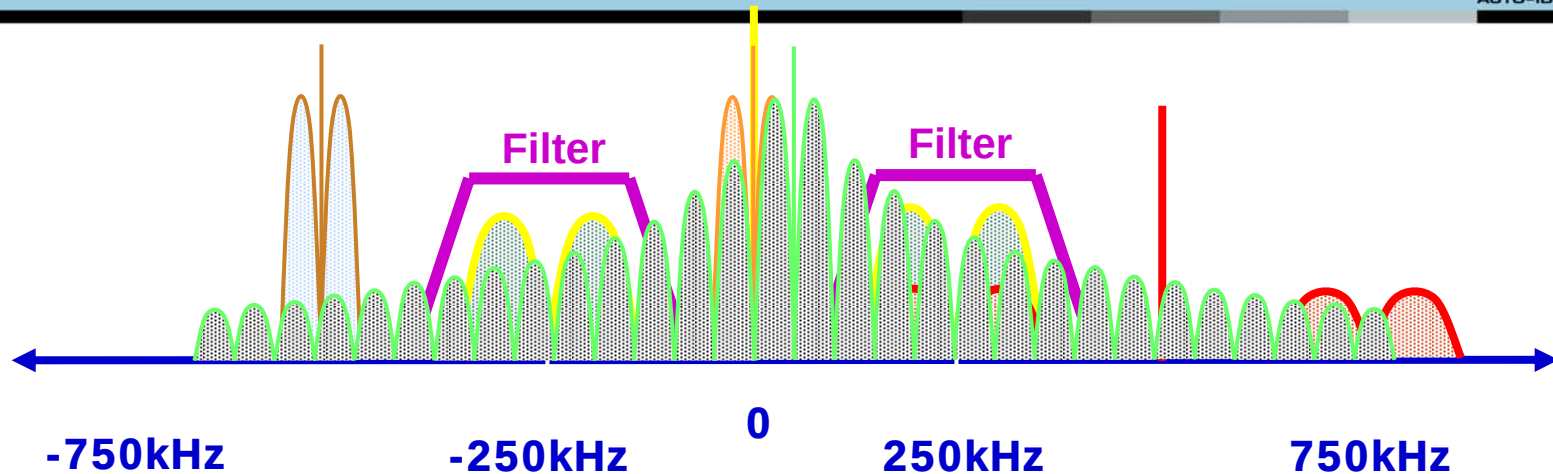
- **More than one readers are needed to cover the working area**
- **Every reader transmit a few watts while tag backscattered power is less than 1 microwatt**
- **If two readers work in the same channel**
 - Signal from the other reader will be much stronger than signal from tags
 - Reader receiver will saturate
- **If readers have to work in different channel, there is not enough bandwidth**

→ Reader to Reader Interference



- One tag may get RF signal from more than one readers
- RF signal added together so it is corrupted
- Tag get confused for more than 1 readers

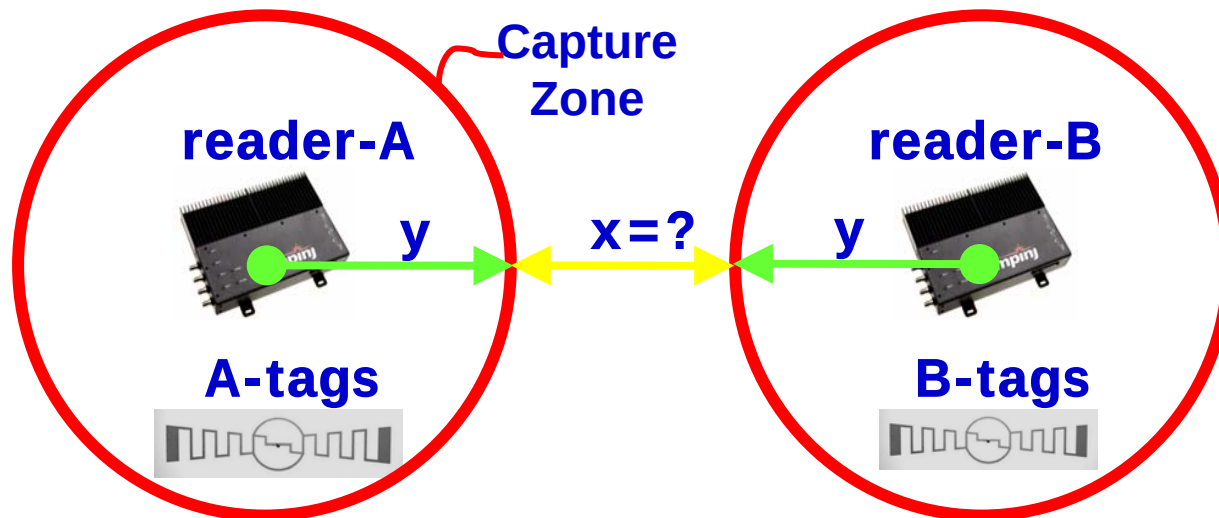




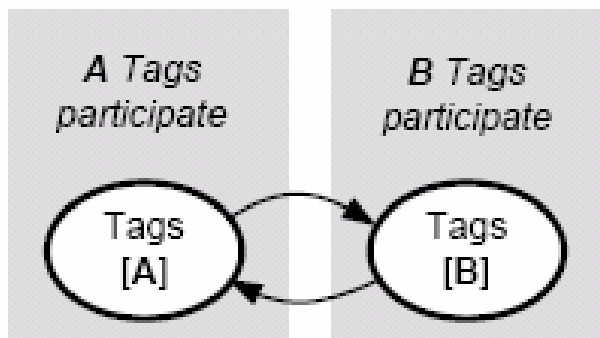
Readers collide with readers but not tags
Readers filter interfering readers from their tag responses

- **Prevents reader—tag interference**
 - Isolates tags and readers in frequency
- **Eliminates need for reader synchronization**
 - Replaces poor-performing listen-before-talk
- **Requires dense-reader certification**
 - One non-dense reader can spoil the whole plan

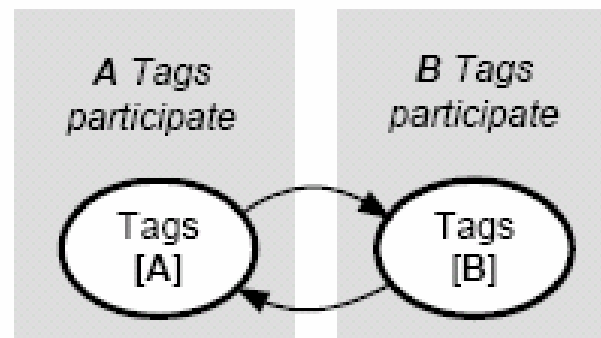
- Readers can detect weak tags as reader and tag use different frequency
- Typical tags are not frequency selective
 - From far-away readers
 - On channel or in different channels
- **Solution: Selective tags according to power strength**



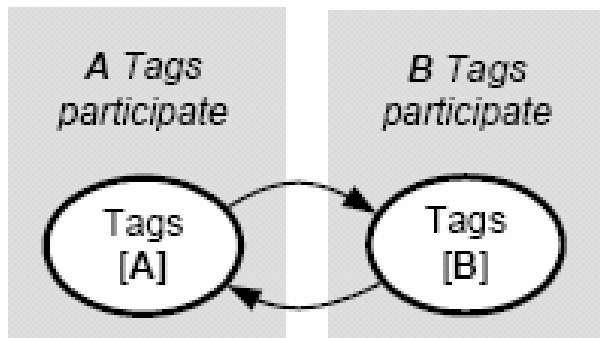
Session S0



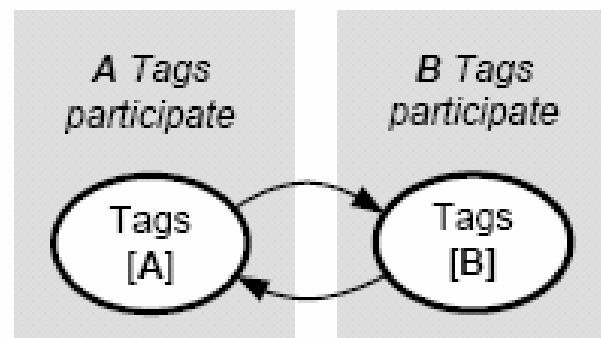
Session S1



Session S2



Session S3





Questions



- **Questions?**
- **Contact:**
 - Hao Min
 - Auto-ID Lab at Fudan
 - Email: hmin@fudan.edu.cn





- This lecture note has been summarized from the research work of Auto-ID Labs and other research institutes all over the world. I can't remember where those slide come from. However, I'd like to thank all professors who create such a good work on those lecture notes. Without those lectures, this slide can't be finished
- Special thanks to Prof. Chris Diorio, who provided me very detailed analysis on Gen2 tags and readers.